Multimedia Signal Processing - 2nd Module

July, 22 2019

1. **Question (8 pts)**
   Consider a continuous source X that emits the following series of symbols

   $$x = \{3,\ 3,\ 9,\ 6,\ 6,\ 0\}.$$

   You are given a quantizer $Q(\cdot)$ that applies the following quantization rule

   $$Q(x) = \left\lfloor \frac{x}{\Delta} \right\rceil \cdot \Delta,$$

   where $\lfloor \cdot \rceil$ rounds to the nearest integer, and $\Delta = 2$ is the quantization step.
   (a) Compute the sequence $\hat{x}_{\text{PCM}}$ obtained by encoding and decoding x using a PCM scheme.
   (b) Compute the mean squared error (MSE) introduced by the PCM.
   (c) Compute the sequence $\hat{x}_{\text{DPCM}}$ obtained by encoding and decoding x using a closed-loop DPCM scheme. Use the following prediction rule

   $$\hat{x}(n) = 1.5 \cdot \hat{x}(n-1),$$

   and use a PCM for the first sample.
   (d) Compute the mean squared error (MSE) introduced by the DPCM.
   (e) Compute the entropy of the sequence $\hat{x}_{\text{DPCM}}$.
   (f) Derive an Huffman code to transmit $\hat{x}_{\text{DPCM}}$.
   (g) Compute the average code length of the derived Huffman code, and compare it with the theoretical bound.

2. **Question (5 pts)**
   Consider a transform coding scheme.
   (a) Sketch a block diagram to explain how transform coding works considering the example of a source that emits blocks of N symbols.
   (b) Considering a source that emits blocks of 2 symbols, explain through an example the geometrical interpretation of transform coding.

3. **Question (5 pts)**
   Consider perceptual audio coding.
   (a) Quickly explain the principle behind it in reference to the acoustic threshold of hearing.
   (b) Explain masking effects in time and frequency domain.

4. **Questions (5 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Consider a video coding scheme, motion vectors are always the same despite the algorithm used to estimate them.
   - [T] [F] Lloyd-Max Quantizer can achieve better performance than the Optimum Uniform Scalar Quantizer.
   - [T] [F] Given two sources X and Y characterized by finite element alphabets,the joint entropy $H(X,Y)$ cannot be smaller than $H(X)$.
   - [T] [F] Given a transform coding scheme, DCT is typically faster to be computed than KLT.
   - [T] [F] In a typical image coding scheme, chrominance and luminance components are equally important.

5. **Question (Matlab - 10pts)**

   Load the default Matlab image `cameraman` using the command `imread('cameraman.tif')`.

   (a) Display the image.

   (b) Encode the image using a DCT-based transform coding scheme using 2×2 pixel blocks and the quantization matrix

   $$\mathbf{Q} = \begin{bmatrix} 10, & 20 \\ 20, & 30 \end{bmatrix}, \tag{1}$$

   where each entry represents the quantization step to use for a different DCT coefficient.

   (c) Compute the rate (bit/pixel) of the used coding scheme in terms of entropy.

   (d) Decode the encoded image and display it.

   (e) Compute the PSNR against the original picture.